Spectral scale selection principle in TDA

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Abstract

We perform spectral analysis of a similarity matrix built from certain functionals on abstract simplicial complexes. The Perron eigenvector of the similarity matrix on the chain of complexes provides a rigorous, robust, and lossless way to choose one or few "best" scales from a multiscale analysis accounting for interactions across different scales. Connection to stationary distributions of some homogeneous irreducible Markov chains on scales as states is stressed.

1 Why a single scale matters?

Topolgical data analysis (TDA) offers a perspective by emphasizing qualitative shape features—connected components, cycles, and higher-dimensional holes—with invariants that are stable to small perturbations [10, 36]. Simplicial complexes built on point clouds according to some rule involving the distance between points (such as Vietoris–Rips (VR) or Čech (Č) compexes) and persistent homology are central constructions that permit multi-scale analysis [16].

Persistent homology is, by design, scale-agnostic: its central output, the persistence diagram, records topological features across *all* filtration scales simultaneously and thereby avoids committing to any single resolution. This is a genuine strength, and no scale-selection procedure can or should replace the full persistence diagram as a summary of multiscale structure. Nevertheless, a surprising number of problems in and around TDA reduce — explicitly or implicitly — to the choice of *one* filtration parameter, and doing so in a theoretically justified way remains a longstanding methodological gap. We list four representative settings in which a topologically justified choice of scale is essential.

Reconstruction and meshing from point clouds. In geometric reconstruction from noisy samples of a manifold $M \subset \mathbb{R}^d$ (shape reconstruction, surface meshing, non-manifold feature detection), the output is a single simplicial complex whose homotopy type should recover that of M. Classical results of Niyogi–Smale–Weinberger [29] and the subsequent reach-based theory [1] guarantee homotopy-type recovery from the VR or Č complex at scale r *only* for r in a specific interval determined by the reach of M and the sampling density. The persistence diagram contains the right interval but does not, by itself, identify it; in applications the reconstruction algorithm must commit to a single r, and a wrong choice produces a mesh with spurious components or filled-in holes. A data-driven scale-selection procedure is not a replacement for persistent homology here but rather its operational completion.

Topology-aware clustering and graph construction. Unsupervised clustering methods that rely on neighbourhood graphs — mean shift, Louvain, HDBSCAN, diffusion maps, Laplacian eigenmaps, UMAP [4,9,28] — all depend on a single scale parameter (k in k-NN, bandwidth

ε in ε-neighbourhood graphs, perplexity in t-SNE). These parameters are typically chosen by heuristics (elbow rule, stability, cross-validation on a supervised surrogate), none of which use the topological structure of the data itself. Yet the clustering quality is known to depend sharply on the scale: too small yields fragmented clusters, too large merges genuine clusters. When the point cloud is drawn from a mixture with multiple characteristic length scales — hierarchical cluster structure, fractal dusts, hyperuniform patterns — no single heuristic performs uniformly well. A topologically-informed selection of scale, grounded in the spectral content of a scale similarity matrix, addresses this gap directly.

Statistical inference for point processes. In the now-classical pipeline of combining topological summaries with spatial-statistics machinery [6, 23], persistent Betti numbers are evaluated on a filtration and then aggregated into a test statistic via Monte Carlo envelope or chi-squared procedures. The full persistence diagram is rarely used as the final test statistic; instead, Betti numbers $\beta_k(r)$ *at a specific scale* r (or a small number of scales) enter the inference. The choice of r is typically made by visual inspection or by matching to the pair correlation function, both of which lack topological justification. For hyperuniform and determinantal point processes — where the relevant length scales are emergent and not set by an obvious geometric parameter [24, 35] — this gap is particularly acute. A scale-selection procedure based on the spectral analysis of scale similarity matrices may provide the missing step and permit downstream tests of homogeneity, goodness-of-fit, and anomaly detection to operate on a well-motivated single complex.

Materials and physical systems with intrinsic length scales. In materials-science applications of TDA — porous media, fibre composites, granular packings, glassy matter [20, 31, 32] — the relevant topological questions (connectivity of the pore space, percolation threshold, distribution of bottleneck sizes) have answers that *do* depend on a specific length scale, namely the characteristic pore size, fibre diameter, or correlation length of the microstructure. The persistence diagram contains this information, but the downstream physical analysis (permeability computation, mechanical simulation, effective-medium modelling) operates on a single complex. Identifying the topologically critical scale from the data itself — rather than imposing it from an external model — turns a multiscale topological summary into a calibrated microstructure descriptor.

The above four settings illustrate that critical-scale selection is not an alternative to persistent homology but its natural operational companion. What is missing, and what this paper aims to provide, is a rigorous spectral mechanism — analogous to the role of eigenvalue decomposition in classical PCA — for identifying this scale from the data itself.

In this work we develop a pipeline that couples a chain of N simplicial complexes, possibly built on a data point cloud in $\mathbb{R}^d$ (VR, $\check{\text{C}}$ or alike), with a chosen functional on simplicial complexes to form a spectral selection principle for the "right" scale. The key idea is to introduce a kind of a *divergence* on simplicial complexes, which is inspired by the recent preprint [33]. The role of their algebraic entropy measure on a topological space is played in our case by any topological or combinatorial functional f of interest, see Sections 2.2, 2.3 and 4 for examples. This enables us to track how the f–complexity changes with the scale parameter ε and to summarize cross-scale interactions in a nonnegative symmetric matrix Σ_N with entries being the "divergencies" of simplicial complexes on different scales ε_j. By Frobenius–Perron theorem (see e.g. [5, Theorem 2.1.1] or [21]), the maximal eigenvalue of Σ_N is positive with the corresponding positive eigenvector v_N whose position of the maximal coordinate reveals the critical scale $\varepsilon_{\text{crit}}$.

The paper is organized as follows: Section 2 offers some preliminaries on simplicial complexes as well as functionals of interest in TDA. Our core spectral scale selection method on simpicial complexes is described in Section 3 including both monotone and non–monotone functionals f.

There, a connection of our methodology to extremal probabilities of steady states of discrete homogeneous Markov chains on scales as state space is stressed. Section 4 continues with some toy examples illustrating our theory. Numerical experiments with alpha, Čech and VR complexes are given in Section 5 followed by a discussion.

2 Preliminaries from computational topology

2.1 Simplicial Complexes

A (finite, abstract) *simplicial complex* K on a vertex set V is a family of finite subsets of V that is closed under taking subsets: if $\sigma \in K$ and $\tau \subseteq \sigma$, then $\tau \in K$. Elements $\sigma \in K$ are called *faces* (or *simplices*). The *dimension* of a nonempty face σ is $\dim \sigma := |\sigma| - 1$, where $|\sigma|$ is the cardinality of σ; by convention the empty face $\varnothing$ has dimension -1. The *dimension* of K is $\dim K := \max\{\dim \sigma : \sigma \in K\}$, possibly -1 if $K = \{\varnothing\}$. A simplex $\sigma \subseteq K$ is called a *facet* of K if there is no $\tau \in K$, $\tau \neq \sigma$ such that $\sigma \subset \tau$. The *dimension* of K is $\dim(K) = \max\{\dim(\sigma) : \sigma \in K\}$. A 0-simplex (singleton set) is called a *vertex*. A 1-simplex (two-element set) is called an *edge*. In general, a k-simplex is a $(k+1)$-element set, representing a k-dimensional face.

The *geometric realization* $\tilde{K}$ of an abstract simplicial complex K is a tolopogical space obtained by embedding each k-simplex $\sigma = \{v_0, v_1, \ldots, v_k\}$ as the convex hull of $(k+1)$ affinely independent points in some Euclidean space $\mathbb{R}^d$, and taking the set union of all such geometric simplices with appropriate identifications along their faces.

Let $\oplus$ denote any operation on simplicial complexes mapping a pair of them onto a new one. In our later examples in Sections 3 and 5, we often take $\oplus$ to be the *union* $\cup$ (or *join* $*$) operation, i.e. $K_1 \cup K_2 := \{\sigma \mid \sigma \in K_1 \text{ or } \sigma \in K_2\}$ or $K_1 * K_2 = \left\{\sigma \cup \tau \;\middle|\; \sigma \in K_1,\ \tau \in K_2\right\}$. However, depending on the context, any other binary operation (such as wedge or different types of products) is admissible as well.

A *partially ordered set* or *poset* is a pair $(P, \leq)$ where P is a set and $\leq$ is a partial order on P. The *Hasse diagram* of a finite poset $(P, \leq)$ is a directed acyclic graph where:

1. Vertices represent elements of P,
2. There is an edge from x to y if $x \leq y$ and there is no $z \in P$ with $x \leq z \leq y$,
3. Vertices are positioned so that $x \leq y$ implies x is drawn below y,
4. Arrows are omitted since the direction is indicated by position.

The *Hasse diagram* or *face poset* of a simplicial complex K is the poset of simplices of K ordered by their inclusion.

Let X be a topological space. An *open cover* of X is a collection $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ of open subsets of X such that $X \subseteq \bigcup_{\alpha \in A} U_\alpha$.

Definition 1. *Let $\mathcal{U} = \{U_0, U_1, \ldots, U_n\}$ be a finite open cover of a topological space X. The abstract* Čech complex *(or* nerve*) $\mathcal{N}(\mathcal{U})$ of the cover $\mathcal{U}$ is the abstract simplicial complex defined by:*

$$\mathcal{N}(\mathcal{U}) = \left\{ \sigma \subseteq \{0, 1, \ldots, n\} : \sigma \neq \emptyset \text{ and } \bigcap_{i \in \sigma} U_i \neq \emptyset \right\}.$$

In other words, a subset $\sigma = \{i_0, i_1, \ldots, i_k\} \subseteq \{0, 1, \ldots, n\}$ forms a k-simplex in $\mathcal{N}(\mathcal{U})$ if and only if the intersection $U_{i_0} \cap U_{i_1} \cap \cdots \cap U_{i_k}$ is non-empty.

Let (X, ρ) be a metric space and $\varepsilon > 0$. For a finite subset $S = \{x_1, x_2, \ldots, x_n\} \subseteq X$, the *Čech complex* $\check{C}_\varepsilon(S)$ is the nerve of the cover $\mathcal{U}_\varepsilon = \{B_\varepsilon(x_i)\}_{i=1}^n$, where $B_\varepsilon(x_i) = \{y \in X : \rho(x_i, y) \leq \varepsilon\}$ is the closed ball of radius ε centered at x_i. The *VR complex* $VR_\varepsilon(S)$

Complex Type	Construction Time	Memory For Storage
Vietoris-Rips	$O(n^2)$ distances	$O(n^{d+1})$
Čech	$O(n^{d+1})$ intersections	$O(n^{d+1})$
Alpha	$O(n^{\lceil d/2 \rceil})$	$O(n^{\lceil d/2 \rceil})$

Table 1: Computational and storage complexity for different simplicial complexes of an n-point set $S \subset \mathbb{R}^d$.

is the abstract simplicial complex whose k-simplices correspond to subsets of $k+1$ points with pairwise distances at most ε. The *Delaunay complex* $\mathrm{Del}(S)$ is the nerve of the Voronoi diagram of S. A simplex $\sigma \subseteq S$ belongs to $\mathrm{Del}(S)$ if and only if the corresponding Voronoi cells have a common intersection. The0 α complex provides a subcomplex of the Delaunay triangulation and is defined as

$$\alpha_\varepsilon(S) = \{\sigma \in \mathrm{Del}(S) : r(\sigma) \leq \varepsilon\}$$

where $r(\sigma)$ is the circumradius of the simplex σ.

As ε increases, all these complexes form an increasing filtration, i.e., it holds

$$VR_{\varepsilon_1}(S) \subseteq VR_{\varepsilon_2}(S), \quad \check{C}_{\varepsilon_1}(S) \subseteq \check{C}_{\varepsilon_2}(S), \quad \alpha_{\varepsilon_1}(S) \subseteq \alpha_{\varepsilon_2}(S), \quad 0 < \varepsilon_1 \leq \varepsilon_2.$$

Moreover, the inclusion $\alpha_\varepsilon(S) \subseteq \check{C}_\varepsilon(S) \subseteq VR_\varepsilon(S) \subseteq \check{C}_{2\varepsilon}(S)$ is valid for all for $\varepsilon > 0$. Each complex type captures different aspects of the data geometry, allowing to test the robustness of our spectral scale selection principle across various topological and combinatorial constructions.

2.2 Topological features of simplicial complexes

2.2.1 Betti numbers

Let K be a finite simplicial (or, more generally, CW-) complex and let $\mathbb{F}$ be a field (typically $\mathbb{F} = \mathbb{Q}$ or $\mathbb{F} = \mathbb{F}_p$ for a prime p). The simplicial chain complex of K with coefficients in $\mathbb{F}$,

$$\cdots \longrightarrow C_{j+1}(K;\mathbb{F}) \xrightarrow{\partial_{j+1}} C_j(K;\mathbb{F}) \xrightarrow{\partial_j} C_{j-1}(K;\mathbb{F}) \longrightarrow \cdots,$$

has $C_j(K;\mathbb{F})$ equal to the $\mathbb{F}$-vector space freely generated by the j-dimensional simplices of K, and boundary operators ∂_j defined combinatorially in the standard way, see [7]. The j-th simplicial homology group with field coefficients is the quotient vector space

$$H_j(K;\mathbb{F}) \;:=\; \frac{\ker \partial_j}{\operatorname{im} \partial_{j+1}}, \qquad j \geq 0.$$

The j-th *Betti number* of K with respect to the field $\mathbb{F}$ is

$$\beta_j(K;\mathbb{F}) \;:=\; \dim_{\mathbb{F}} H_j(K;\mathbb{F}), \qquad j \geq 0. \tag{1}$$

When the coefficient field is fixed (most commonly $\mathbb{F} = \mathbb{Q}$ or $\mathbb{F} = \mathbb{F}_2$), one writes simply $\beta_j(K)$. By convention, $\beta_j(K) = 0$ for $j > \dim K$, since $C_j(K;\mathbb{F})$ vanishes in those degrees.

The Betti numbers admit the following geometric interpretation:

- $\beta_0(K)$ is the number of connected components of K;
- $\beta_1(K)$ is the number of independent one-dimensional cycles ("loops" or "handles");
- $\beta_2(K)$ is the number of independent two-dimensional voids ("cavities");
- more generally, $\beta_j(K)$ counts the number of linearly independent j-dimensional "holes" of K.

The alternating sum of Betti numbers recovers the Euler characteristic of K:

$$\chi(K) \;=\; \sum_{j=0}^{\dim K} (-1)^j \, \beta_j(K).$$

The use of Betti numbers β_k in TDA is classical. Except for the number of connected components β_0, they are not subadditive w.r.t. union operation.

2.2.2 Cup length

The *cup length* of a simplicial complex K is a topological invariant derived from the multiplicative structure of its cohomology ring. It measures the longest non-trivial chain of cup products of positive-degree cohomology classes.

Let K be a finite simplicial complex and let R be a coefficient ring (typically $R = \mathbb{Z}$, $\mathbb{Q}$, or a field $\mathbb{F}_p$). The simplicial cohomology of K with coefficients in R,

$$H^*(K;R) \;=\; \bigoplus_{n\geq 0} H^n(K;R),$$

carries a graded-commutative multiplication

$$\smile\colon H^p(K;R) \otimes H^q(K;R) \;\longrightarrow\; H^{p+q}(K;R),$$

called the *cup product*, defined at the cochain level by the Alexander–Whitney formula [18] Recall that by the Universal Coefficient Theorem (see [7]), when R is a field, the following isomorphism holds $H^n(K;R) \cong H_n(K;R)$, so from a TDA point of view, we are still working with homology, but with the multiplication added. The pair $(H^*(K;R), \smile)$ is a graded-commutative ring with unit $1 \in H^0(K;R)$, called the *cohomology ring* of K.

Let $\widetilde{H}^+(K;R) := \bigoplus_{n\geq 1} H^n(K;R)$ denote the reduced (positive-degree) part of the cohomology, i.e. the kernel of the augmentation map $H^*(K;R) \to R$.

Definition. The *cup length* of K with coefficients in R, denoted $\mathrm{cup}_R(K)$ (or simply $\mathrm{cup}(K)$ when the coefficient ring is understood), is

$$\mathrm{cup}_R(K) \;=\; \max\bigl\{\, k \geq 0 \;\bigm|\; \exists\, \alpha_1, \ldots, \alpha_k \in \widetilde{H}^+(K;R) \text{ with } \alpha_1 \smile \alpha_2 \smile \cdots \smile \alpha_k \neq 0 \,\bigr\}. \tag{2}$$

By convention, $\mathrm{cup}_R(K) = 0$ when no such non-trivial product exists, i.e. when every positive-degree class is nilpotent of order one and the cohomology ring of K is trivial in positive degrees.

Equivalent formulation. The cup length equals the nilpotency index of the augmentation ideal of the cohomology ring, minus one:

$$\mathrm{cup}_R(K) \;=\; \max\bigl\{\, k \geq 0 \;\bigm|\; \widetilde{H}^+(K;R)^{\smile k} \neq 0 \,\bigr\},$$

where $I^{\smile k}$ denotes the k-th cup-power of an ideal I.

Standard examples. The cup length of a few classical topological spaces is collected in Table 2.

Complex K	$\mathrm{cup}_R(K)$	**Reason**
Point K	0	$\widetilde{H}^+(K) = 0$
S^n $(n \geq 1)$	1	$H^*(S^n) = R[\alpha]/(\alpha^2)$, $\deg \alpha = n$
$\mathbb{C}P^n$	n	$H^*(\mathbb{C}P^n) = R[\alpha]/(\alpha^{n+1})$, $\deg \alpha = 2$
$\mathbb{R}P^n$ ($\mathbb{F}_2$-coeff.)	n	$H^*(\mathbb{R}P^n; \mathbb{F}_2) = \mathbb{F}_2[\alpha]/(\alpha^{n+1})$, $\deg \alpha = 1$
$T^n = (S^1)^n$	n	exterior algebra on n degree-1 generators
Contractible K	0	trivial reduced cohomology
$S^m \vee S^n$	1	products of generators lie in different summands
$S^m \times S^n$	2	the product class $\alpha \smile \beta \neq 0$, $\deg \alpha = m$, $\deg \beta = n$

Table 2: Cup length of standard simplicial complexes (coefficient ring as indicated; default $R = \mathbb{Z}$).

Key properties. The cup length satisfies the following properties [7, 11, 18]:

1. **Homotopy invariance.** $\mathrm{cup}_R(K)$ depends only on the homotopy type of the geometric realisation $|K|$, not on the simplicial structure: two homotopy-equivalent complexes have the same cup length.
2. **Coefficient dependence.** cup_R depends on the coefficient ring. For instance, $\mathrm{cup}_{\mathbb{Z}}(\mathbb{R}P^n) = \lfloor \frac{n}{2} \rfloor$ (since the integral cohomology is mostly torsion), whereas $\mathrm{cup}_{\mathbb{F}_2}(\mathbb{R}P^n) = n$.
3. **Lower bound for the number of critical points.** For a smooth function $f\colon M \to \mathbb{R}$ on a closed manifold M, f has at least $\mathrm{cup}_R(M) + 1$ critical points.
4. **Behaviour under products.**
$$\mathrm{cup}_R(K \times L) \;\geq\; \mathrm{cup}_R(K) + \mathrm{cup}_R(L),$$
with equality when $H^*(K \times L; R) \cong H^*(K; R) \otimes H^*(L; R)$ as graded rings (Künneth, e.g. when R is a field).
5. **Behaviour under wedge.** Cup products of classes in different summands of a wedge vanish, so
$$\mathrm{cup}_R(K \vee L) \;=\; \max\bigl(\mathrm{cup}_R(K),\, \mathrm{cup}_R(L)\bigr).$$
$$\mathrm{cup}_R(K_1 \cup K_2) \;\leq\; \mathrm{cup}_R(K_1) + \mathrm{cup}_R(K_2) + 1. \tag{3}$$
6. **Bounded by dimension.**
$$\mathrm{cup}_R(K) \;\leq\; \dim K,$$
since a non-zero k-fold cup product of classes of total degree at least k requires the existence of cohomology in degree at least k.

Computation. Cup length is computable from a presentation of the cohomology ring. The standard algorithm is:

1. Compute the cohomology groups $H^n(K; R)$ from the simplicial cochain complex.
2. Compute the cup-product maps $H^p(K; R) \otimes H^q(K; R) \to H^{p+q}(K; R)$ via the Alexander–Whitney formula at the cochain level, then descend to cohomology.
3. Find the maximum k such that some k-fold cup product of positive-degree generators is non-zero.

For K of dimension d with f-vector $(f_0, \ldots, f_d)$, the computation requires $O\bigl(\sum_n f_n^2\bigr)$ work to set up the cup products plus additional linear-algebra work to determine non-vanishing of products.

2.3 Combinatorial features of simplicial complexes

2.3.1 f-vectors and their scalar product

For $-1 \le i \le d := \dim K$, denote by $f_i(K)$ the number of i-dimensional faces of K (with $f_{-1}(K) := 1$ counting the empty face). The sequence $\big(f_{-1}(K), f_0(K), \dots, f_d(K)\big)$ is the *f-vector* of K.

Definition 2 (f-polynomial)**.** *Let K be a finite simplicial complex of dimension d. The f-*polynomial *is*

$$F_K(t) := \sum_{i=-1}^{d} f_i(K)\, t^{\,i+1} = 1 + f_0(K)t + f_1(K)t^2 + \ldots + f_d(K)t^{d+1}.$$

Remark 1 (Interpretation for data)**.** *For a complex built from data (e.g. a Vietoris–Rips complex $VR(X,\varepsilon)$), f_0 is the number of points, f_1 the number of edges, f_2 the number of filled triangles, etc. The f-vector summarizes* combinatorial complexity *at a given scale ε. Tracking f_i as ε changes provides a low-cost diagnostic complementary to persistent homology.*

Example 1 (Simplex and its boundary)**.** *Let Δ^n denote the n-simplex on $n+1$ vertices. Then*

$$f_i(\Delta^n) = \binom{n+1}{i+1} \quad \text{for } -1 \le i \le n,$$

so $F_{\Delta^n}(t) = \sum_{i=-1}^{n} \binom{n+1}{i+1} t^{\,i+1} = (1+t)^{n+1}$. Its boundary $\partial\Delta^n$ has

$$f_i(\partial\Delta^n) = \binom{n+1}{i+1} \quad \text{for } -1 \le i \le n-1, \qquad f_n(\partial\Delta^n) = 0.$$

Example 2 (Graphs as 1-complexes)**.** *If K is a graph with no loops or multiple edges viewed as a 1-dimensional complex, then $f(K) = (1, |V|, |E|)$ and the Euler characteristic is $\chi(K) = |V| - |E|$.*

Let us recall the basic properties of f-vectors.

Proposition 1 (Monotonicity and bounds)**.** *Let K be finite of dimension d.*

1. *If $L \subseteq K$ is a subcomplex then $f_i(L) \le f_i(K)$ for all i.*
2. *(Trivial upper bounds) Writing $n = f_0(K)$, for each $i \ge 0$ one has $f_i(K) \le \binom{n}{i+1}$, with equality for all $i \le d$ iff K is the full (d)-skeleton on n vertices, i.e. $K = \{\sigma \subseteq V : |\sigma| \le d+1\}$ for $V = \{1, \dots, n\}$.*

Proposition 2 (Euler characteristic)**.** *The (reduced) Euler characteristic satisfies*

$$\tilde{\chi}(K) = \sum_{i=-1}^{d} (-1)^i\, f_i(K) \qquad \big(\text{and } \chi(K) = \sum_{i=0}^{d} (-1)^i\, f_i(K)\big).$$

Proposition 3 (Additivity under disjoint union)**.** *If K and L are complexes on disjoint vertex sets, then*

$$f_{-1}(K \sqcup L) = 1, \qquad f_i(K \sqcup L) = f_i(K) + f_i(L) \quad \text{for } i \ge 0,$$

equivalently $F_{K\sqcup L}(t) = F_K(t) + F_L(t) - 1$.

Proposition 4 (Subadditivity under unions)**.** *For any simplicial complexes K and L on the same vertex set and any $i \ge -1$,*

$$f_i(K \cup L) = f_i(K) + f_i(L) - f_i(K \cap L) \;\le\; f_i(K) + f_i(L).$$

In particular, equality $f_i(K \cup L) = f_i(K) + f_i(L)$ holds iff K and L share no i-faces.

Proposition 5 (Join)**.** *For the join $K * L$ (on disjoint vertex sets) one has*

$$F_{K*L}(t) = F_K(t)\, F_L(t),$$

equivalently

$$f_r(K * L) = \sum_{i+j=r-1} f_i(K)\, f_j(L) \qquad (r \geq -1).$$

Remark 2 (Face numbers vs. topology)**.** *The f-vector encodes counts, not homology. Many non-isomorphic complexes can share the same f-vector. Nevertheless, together with orientations/boundaries, f-vectors control the sizes of boundary matrices used to compute homology, which impacts memory/time in TDA pipelines.*

Given an n–point set S and a scale parameter ε, the full Vietoris–Rips complex $K_\varepsilon(S)$ of point cloud S has

$$f_i(K_\varepsilon(S)) \leq \binom{n}{i+1} \quad \textit{and typically grows rapidly with } i.$$

In practice one could truncate *to a maximum dimension d (e.g. $d \in \{2, 3, 4\}$) to control memory. The f-vector up to d gives a quick sanity check on whether a computation is feasible before constructing boundary matrices.*

2.3.2 Counting combinatorial functionals

Any function of the form

$$f(K) = \sum_{T \in \mathcal{T}} a_T \cdot \#\{\text{occurrences of } T \subseteq K\},$$

where $\mathcal{T}$ is a finite set of template subcomplexes and $a_T \geq 0$ are integers, is nonnegative, integer-valued, and subadditive (since counting occurrences is subadditive with respect to union operation). The most prominent functions f of this type are e.g. numbers of faces f_i of a certain dimension $i \in \{0, \ldots, \dim K\}$ and numbers of specific subcomplex patterns (e.g., "simplets", cf. [3]).

Let us give a number of nonnegative combinatorial functions on a simplicial complex $K = K(S)$ constructed on vertices $V(K) := S$ that are subadditive and, moreover, monotone with respect to inclusion. Denote by $\mathcal{F}(K)$ the set of all facets of K, i.e., simplexes of maximal dimension in K.

Chain counts The *chain count of length* $r \in \mathbb{N}$ is the number of strictly increasing chains of simplices of length r in K:

$$c_r(K) = |\Big\{(\sigma_0 \subsetneq \sigma_1 \subsetneq \cdots \subsetneq \sigma_r) : \ \sigma_i \in K\Big\}|, \quad r \geq 1.$$

Actually, $c_r(K)$ is equal to the number of r faces of the barycentric subdivision of K, see [18].

Because every chain of length r must be entirely contained within some maximal simplex of K, we can compute $c_r(K)$ directly from the f-vector of K.

Let $f_j(K)$ be the number of j-dimensional simplices in K. Then:

$$c_r(K) = \sum_{j=r}^{\dim K} f_j(K) \cdot c_r(\Delta^j)$$

where $c_r(\Delta^j)$ is the number of chains of length r in a single j-dimensional simplex.

There is a beautiful closed-form formula for $c_r(\Delta^j)$ using **Stirling numbers of the second kind**, denoted $S(n, k)$ [8] (which count the number of ways to partition a set of n elements into k non-empty subsets):

$$c_r(\Delta^j) = (r+1)! \cdot S(j+2, r+2)$$

Maximal coface-degree For a simplicial complex K and a face $\tau \in K$, the *t-coface degree* of τ is defined by

$$\deg_t(\tau) = |\{\sigma \in K : \tau \subseteq \sigma,\ \dim(\sigma) = t\}|\,.$$

The *maximum t-coface degree* of K is

$$\Delta_t(K) = \max_{\substack{\tau \in K \\ 0 \le \dim(\tau) < t}} |\{\sigma \in K : \tau \subseteq \sigma,\ \dim(\sigma) = t\}|\,.$$

Vertex cover number The *vertex cover number*, also known as the *transversal number*, of K is

$$\mathrm{vc}(K) = \min\Big\{|C| : C \subseteq V(K),\ C \cap \sigma \neq \varnothing \text{ for every facet } \sigma \text{ of } K\Big\}.$$

Facet matching number Let K be a simplicial complex with set of facets $\mathcal{F}(K)$. A collection $M \subseteq \mathcal{F}(K)$ is called a *facet matching* if its elements are pairwise disjoint. The *facet matching number* of K is

$$\nu_{\mathcal{F}}(K) = \max\Big\{|M| : M \subseteq \mathcal{F}(K),\ \sigma \cap \sigma' = \varnothing \text{ for all distinct } \sigma, \sigma' \in M\Big\}.$$

Total vertex deficiency Another combinatorial characteristic of K is given by the so-called *total vertex deficiency*:

$$\mathrm{VD}(K) := \sum_{x \in S} vd(x),$$

where S is the set of all vertices of K (faces of zero dimension) and

$$vd(x) := d - \max\{\dim(\sigma) \mid x \in \sigma, \sigma \in K\}$$

is the *vertex deficiency* of a point $x \in S$. Here d is the dimension of K.

Chromatic number Let K be a finite simplicial complex with vertex set $V = V(K)$ and let $K^{(m)}$ denote the *m-skeleton* of K, i.e. the subcomplex consisting of all simplices of dimension at most m. For $m = 1$, $K^{(1)}$ is the underlying graph of K.

A *proper k-colouring* of $K^{(m)}$ is a map

$$c\colon V \to \{1, 2, \ldots, k\}$$

such that on every simplex of $K^{(m)}$ of dimension at least 1 (every edge, triangle, ..., up to m-simplex), the colouring c is not constant — equivalently, no m'-simplex with $m' \le m$ has all its vertices receiving the same colour. The most natural and classical choice is to require non-monochromaticity only on edges, in which case the condition reduces to "no edge of $K^{(1)} \subseteq K^{(m)}$ is monochromatic" — which is the standard graph-theoretic notion.

Define the k-chromatic number of K as

$$CR_{k,m}(K) \ := \ \#\big\{\, c\colon V \to \{1, \ldots, k\} \ \big|\ c \text{ is a proper colouring of } K^{(m)} \,\big\}.$$

2.4 Simplicial complexes assigned to a graph and their applications

There are many ways to assign a simplicial complex to a graph G. Simplicial complexes built from a graph based on a certain combinatorial property have very interesting topological features. In the last decades they were intensively studied in topological combinatorics. Here we introduce some of them and discuss some of their properties in the context of TDA applications.

Definition 3.

- *The* clique complex *(or* flag complex*) of $G = (V, E)$, denoted $X(G)$, is the simplicial complex whose simplices are all finite subsets $\sigma \subseteq V$ such that every pair of vertices in σ is connected by an edge in G. That is,*

$$X(G) = \{\sigma \subseteq V \mid \forall u, v \in \sigma,\, u \neq v \implies \{u, v\} \in E\}.$$

- *The* independence complex *of $G = (V, E)$, denoted $I(G)$, is the simplicial complex whose simplices are all independent (i.e., not connected by an edge) sets of vertices:*

$$I(G) = \{\sigma \subseteq V \mid \forall u, v \in \sigma,\, u \neq v \implies \{u, v\} \notin E\}.$$

- *The* neighborhood complex *of $G = (V, E)$, denoted $N(G)$, is the simplicial complex whose simplices are subsets of vertices having a common neighbor:*

$$N(G) = \{\sigma \subseteq V \mid \exists v \in V \text{ such that } \forall u \in \sigma,\, \{u, v\} \in E\}.$$

- *The* matching complex *of $G = (V, E)$, denoted $M(G)$, is the simplicial complex whose vertices are the edges of G, and whose simplices are sets of pairwise disjoint edges (i.e., matchings):*

$$M(G) = \{\sigma \subseteq E \mid \forall e_1, e_2 \in \sigma,\, e_1 \cap e_2 = \emptyset\}.$$

Let $G = (V, E)$ be a finite simple graph. A subset $S \subseteq V$ is called a *dominating set* if $\forall v \in V$ holds $v \in S$ or $\exists u \in S$ such that $u, v \in E$. In other words, every vertex of the graph is either in S or adjacent to a vertex in S.

Definition 4. *The* dominance complex *of $G = (V, E)$, denoted $D(G)$, is the simplicial complex consisting of complements of dominating sets:*

$$D(G) = \{\sigma \subseteq V \mid V \setminus \sigma \text{ is a dominating set in } G\}.$$

3 Spectral scale selection

Let $f(K)$ be any real–valued functional on abstract simplicial complexes K. Introduce the f-based divergence on simplicial complexes as follows:

Definition 5. *Let K_1, K_2 be simplicial complexes. The* f-based divergence *on K_1, K_2 is given by*

$$\langle K_1, K_2 \rangle_f := f(K_1) + f(K_2) - f(K_1 \oplus K_2).$$

The non–negativity of $\langle \cdot, \cdot \rangle_f$ is assured whenever f is subadditive with respect to operation $\oplus$, i.e., it holds $f(K_1 \oplus K_2) \leq f(K_1) + f(K_2)$ for any simplicial complexes K_1, K_2. However, in some specific cases non-subadditive functionals also yield nonnegativity of f-divergence, as seen below.

Remark 3. *Let K_1, K_2 be simplicial complexes with geometric realizations in $\mathbb{R}^d$, and let $\oplus$ be any operation on them s.t. $K_1 \oplus K_2 \subseteq K_2$ for $K_1 \subseteq K_2$ (e.g., union, intersection, etc.). Assume f to be any nonnegative functional which is monotone w.r.t. inclusion: $f(K_1) \leq f(K_2)$ for any $K_1 \subseteq K_2$. Thus, for $K_1 \subseteq K_2$ it holds*

$$\langle K_1, K_2 \rangle_f \geq f(K_1) + f(K_2) - f(K_2) = f(K_1) \geq 0. \tag{4}$$

In case of the union operation, the monotonicity of f is not needed, since we do have

$$\langle K_1, K_2 \rangle_f = f(K_1) \geq 0 \text{ if } K_1 \subseteq K_2. \tag{5}$$

Now assume that a sequence of simplicial complexes $K_1, \dots, K_N$ and a real–valued functional f on them are given such that $\langle K_i, K_j \rangle_f \geq 0$ for all $i, j = 1, \dots, N$.

Example 3. *Let (X, ρ) be a metric space. For a given finite data point set $S = \{x_1, x_2, \dots, x_n\} \subseteq X$, let $diam(S) := \max_{x,y \in S} \rho(x, y)$ be the diameter of S. Consider an increasing filtration $\{K_\varepsilon\}$ of finite abstract simplical complexes $K_\varepsilon(S)$ built upon S using a scale parameter $\varepsilon \in [0, M]$, where $M = \frac{diam(S)}{2}$. For instance, we may set $K_\varepsilon(S)$ to be equal to $\alpha_\varepsilon(S)$, $\check{C}_\varepsilon(S)$ or $VR_\varepsilon(S)$. Fix an $N \in \mathbb{N}$ to be the number of scales. For a grid $\{\varepsilon_i\}_{i=1}^N$ of filtration parameters (e.g., $\varepsilon_j = \frac{j \cdot M}{N}$, $j = 1, \dots, N$), we put $K_i := K_{\varepsilon_i}(S)$ for any i, $\oplus = \cup$ and f to be any subadditive functional w.r.t the union operation. If $X = \mathbb{R}^d$, we may drop subadditivity and require just nonnegativeness of f, cf. Remark 3.*

Definition 6. *The triple $(f, \{K_i\}, \oplus)$ is said to be spectrally optimizable if it generates a finite irreducible discrete time Markov chain on the state space $\{1, \dots, N\}$ with transition probability matrix $Q = (q_{ij})$ and stationary distribution $\pi^Q = (\pi_1^Q, \dots, \pi_N^Q)$.*

define optimality of scale choice for the pair (f, K_i) via inverse Frobenius-Perron idea

3.1 Scale Similarity Matrix Construction and Eigenvalue Analysis

Define the symmetric nonnegative matrix $\Sigma_N := (\Sigma_{ij})_{i,j=1}^N$ with $\Sigma_{ij} := \langle K_i, K_j \rangle_f$, $i, j = 1, \dots, N$, which will be called a *scale similarity matrix*. It is evidently a Gram matrix

$$\Sigma_N = \begin{pmatrix} \langle K_1, K_1 \rangle_f & \langle K_1, K_2 \rangle_f & \langle K_1, K_3 \rangle_f & \cdots & \langle K_1, K_N \rangle_f \\ \langle K_2, K_1 \rangle_f & \langle K_2, K_2 \rangle_f & \langle K_2, K_3 \rangle_f & \cdots & \langle K_2, K_N \rangle_f \\ \langle K_3, K_1 \rangle_f & \langle K_3, K_2 \rangle_f & \langle K_3, K_3 \rangle_f & \cdots & \langle K_3, K_N \rangle_f \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \langle K_N, K_1 \rangle_f & \langle K_N, K_2 \rangle_f & \langle K_N, K_3 \rangle_f & \cdots & \langle K_N, K_N \rangle_f \end{pmatrix}.$$

Assume that the matrix Σ_N is irreducible, i.e. for all $i, j = 1, \dots, N$ there exists a natural p such that $(\Sigma_N^p)_{ij}$ is positive. To explain this, associate to Σ_N an undirected graph G_{Σ_N} on the vertex set $\{1, \dots, N\}$, with an edge $i \sim j$ whenever $(\Sigma_N)_{ij} > 0$. A positive diagonal entry $(\Sigma_N)_{ii} > 0$ is interpreted as a self-loop. Then, Σ_N is irreducible if and only if the graph G_{Σ_N} is connected.

give a reference

Let λ_j, $j = 1, \dots, N$ be eigenvalues of Σ_N enumerated in ascending order of their absolute values: $|\lambda_1| \leq |\lambda_2| \leq \dots \leq |\lambda_{N-1}| \leq |\lambda_N|$, and let $v_j = (v_{j1}, \dots, v_{Nj})^\top$ be the corresponding orthonormal eigenvectors. Then, it holds the eigenvalue decomposition

$$\Sigma_N = \sum_{j=1}^N \lambda_j v_j v_j^\top. \tag{6}$$

give conditions of filtration and f to assure this

By the Frobenius-Perron theorem [5, Theorem 2.1.4], its maximal eigenvalue $\lambda_N > |\lambda_{N-1}| \geq 0$ with corresponding eigenvector $v_N = (v_{N1}, \dots, v_{NN})^\top$ such that $v_{Nj} > 0$ for all j satisfy

$$\lim_{m \to \infty} \frac{(\Sigma_N)^m}{(\lambda_N)^m} = v_N v_N^\top.$$

Consider the order statistics $v_{j(1)} \leq \dots \leq v_{j(N)}$ of the coordinates of v_j, and take the corresponding index values $k_{j1}, \dots, k_{jN}$ such that $v_{k_{ji}} = v_{j(i)}$, $i = 1, \dots, N$. In case

$$v_{j(1)} < \dots < v_{j(N)} \tag{7}$$

these index values are unique. If not, let

$$v_{j(1)} = \dots = v_{j(m_1)} < v_{j(m_1+1)} = \dots = v_{j(m_1+m_2)} < v_{j(m_1+\dots+m_{r_j-1}+1)} = \dots = v_{j(m_1+\dots+m_{r_j})},$$

where $m_1 + \dots + m_{r_j} = N$, $1 \leq r_j \leq N$ and $m_0 = 0$.

We select the *critical scales* set $\varepsilon_{\text{crit}}$ to be $\varepsilon_{\text{crit}} := \{\varepsilon_{k_{Nm_1+\dots+m_{r-1}+1}}, \dots, \varepsilon_{k_{NN}}\}$. It represents scales with most significant topological or combinatorial changes within our filtration.

Remark 4 (Perron–Frobenius properties of the dominant eigenvector)**.** *In many practical settings, the set ε_{crit} is often a singleton. However, the cases where ε_{crit} has more than one element may come across. They are controlled by the following corollaries of Perron–Frobenius theorem [22].*

If $\Pi \in \mathbb{R}^{N\times N}$ is a permutation matrix with $\Pi\Sigma_N\Pi^\top = \Sigma_N$, then $\Pi v_N = v_N$; consequently, $v_{Ni} = v_{Nj}$ whenever i and j lie in the same orbit of the symmetry group of Σ_N (for instance, invariance under reversal of the filtration parameter $j \leftrightarrow N+1-j$). For generic Σ_N with trivial symmetry group, the entries of v_N are pairwise distinct, and the induced ranking (7) is strict. This follows from the observation that for matrices outside a measure-zero algebraic subset of entries, no non-trivial permutation fixes Σ_N.

If Σ_N admits a non-trivial symmetry orbit $\{i_1, \ldots, i_k\}$ on which $v^{(1)}$ is constant, the ranking may be refined using subleading eigenvectors. The second eigenvector v_{N-1} is by construction orthogonal to v_N, and transforms non-trivially under the symmetry group, so its entries differ across the orbit and can serve as a secondary ranking criterion, and so on.

Remark 5. *If the choice of functional f cannot assure the nonnegativity of the matrix Σ_N, its square Σ_N^2 can be used instead in the above spectral analysis. Matrix Σ_N^2 is symmetric nonnegative definite with eigenvalues $0 \le \lambda_1^2 \le \lambda_2^2 \le \ldots \le \lambda_{N-1}^2 < \lambda_N^2$, which allows us to see it as a covariance matrix. The interpretation of its entries $\Sigma_{ij}^{(2)}$, $i, j = 1, \ldots, N$ may be as follows: $\Sigma_{ij}^{(2)}$, $i \ne j$ represents the* correlation *of changes in the combinatorics/topology of simplicial complexes $K_1, \ldots, K_N$ during the transition from complex K_i to K_j. However, the information about negative eigenvalues of Σ_N gets lost.*

give better interpretation for the Perron vector. Compare it to the Perron vector of Σ_N

Example 4. *In the combined setting of Remark 3 and Example 4, let f be any nonnegative functional on simplicial complexes with geometric realization in $\mathbb{R}^d$ equipped with the union operation. By the monotonicity property of filtration $\{K_{\varepsilon_i}(S)\}_{i=1}^N$, one can easily see that*

$$\Sigma_{ij} = f(K_{\varepsilon_{\min(i,j)}}(S)) \ge 0, \quad i, j = 1, \ldots, N.$$

Thus, the scale similarity matrix writes

$$\Sigma_N = \begin{pmatrix} f(K_{\varepsilon_1}(S)) & f(K_{\varepsilon_1}(S)) & f(K_{\varepsilon_1}(S)) & \cdots & f(K_{\varepsilon_1}(S)) \\ f(K_{\varepsilon_1}(S)) & f(K_{\varepsilon_2}(S)) & f(K_{\varepsilon_2}(S)) & \cdots & f(K_{\varepsilon_2}(S)) \\ f(K_{\varepsilon_1}(S)) & f(K_{\varepsilon_2}(S)) & f(K_{\varepsilon_3}(S)) & \cdots & f(K_{\varepsilon_3}(S)) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f(K_{\varepsilon_1}(S)) & f(K_{\varepsilon_2}(S)) & f(K_{\varepsilon_3}(S)) & \cdots & f(K_{\varepsilon_N}(S)) \end{pmatrix}. \tag{8}$$

It is a special case of the so–called min or meet matrices, cf. Section 3.3 for their properties.

If the functional $f(K_\varepsilon(S))$ is monotonically increasing (decreasing, respectively) in $\varepsilon > 0$, then it is possible to show by induction that $v_{N(j)} = v_{Nj}$ (or $v_{N(j)} = v_{NN-j+1}$, respectively), see Proposition 6 in Section 3.3.2. Hence, in order to get a meaningful scale choice for ε that does not merely reflect the monotonicity of f it is recommended to use non–monotone functionals $f(K_\varepsilon(S))$ in data analysis. An alternative way to cope with monotone functionals f is to consider the scale at which the stationary distribution probabilities of an embedded Markov chain for the birth–and–death process on the state space $\{1, \ldots, N\}$ constructed upon a generator related to Σ_N^{-1} attain their minimum, compare Section 3.4.

3.2 Scale Similarity and Markov Chains

The spectral scale choice made above can be motivated by the association of a finite Markov chain to scale similarity matrix Σ_N such that its stationary distribution is proportional to the squared coordinates of the positive eigenvector $v_N = (v_{N1}, \ldots, v_{NN})^\top$.

The following classical construction (Doob's h-Transform) is borrowed from [14,15]. Define a finite homogeneous Markov chain on the state space $\{1,\dots,N\}$ of indices of simplicial complexes K_i by introducing its transition matrix

$$\widetilde{P} = \frac{1}{\lambda_N} D_{v_N}^{-1} \Sigma_N D_{v_N},$$

where $D_{v_N} = \mathrm{diag}(v_{N1},\dots,v_{NN})$. In other words, $\widetilde{P} = (\widetilde{P}_{ij})_{i,j=1}^N$ with

$$\widetilde{P}_{ij} = \frac{\Sigma_{i,j}\, v_{Nj}}{\lambda_N\, v_{Ni}}, \quad i,j = 1,\dots,N. \tag{9}$$

Starting at state i, the chain jumps to state j with probability proportional to $\Sigma_{i,j} \cdot v_{Nj}$ which is biased toward states with large v_{Nj} (the Perron weighting).

The stochastisity of matrix $\widetilde{P}$ is checked immediately by noting that

$$\widetilde{P}\,\mathbf{1} = \frac{1}{\lambda_N} D_{vN}^{-1} \Sigma_N\, v_N = \frac{1}{\lambda_N} D_{vN}^{-1} \lambda_N\, v_N = \mathbf{1},$$

where $\mathbf{1}$ is the vector with all coordinates being equal to ones. This Markov chain is reversible and ergodic with stationary distribution $\tilde{\pi} = (\tilde{\pi}_1,\dots,\tilde{\pi}_N)^\top$, where

$$\tilde{\pi}_i = \frac{v_{Ni}^2}{\|v_N\|_2^2}, \quad \|v_N\|_2^2 = \sum_{k=1}^N v_{Nk}^2, \quad i = 1,\dots,N.$$

To see this, just check that $\tilde{\pi}_i \widetilde{P}_{ij} = \dfrac{v_{Ni}^2}{\|v_N\|_2^2} \cdot \dfrac{\Sigma_{ij}\, v_{Nj}}{\lambda_N\, v_{Ni}} = \dfrac{v_{Ni}\,\Sigma_{ij}\, v_{Nj}}{\lambda_N \|v_N\|_2^2}$, which is symmetric in i,j due to the symmetry of Σ_N.

The eigenvalues of $\widetilde{P}$ are exactly

$$\widetilde{\mu}_k = \frac{\lambda_k}{\lambda_N} \in [-1,1], \qquad k = 1,\dots,N$$

with $1 = \widetilde{\mu}_N > |\widetilde{\mu}_{N-1}| \geq \ldots \geq |\widetilde{\mu}_1|$. Some eigenvalues of $\widetilde{P}$ lie in $(-1,0)$ — these correspond to *oscillatory modes* in the chain's relaxation, where the chain alternates before settling to equilibrium. The speed of convergence to the stationary regime after n steps is exponential with the upper bound

$$\big\|\widetilde{P}^n(i,\cdot) - \tilde{\pi}\big\|_{\mathrm{TV}} \;\leq\; C|\widetilde{\mu}_2|^n,$$

where $\|\cdot\|_{\mathrm{TV}}$ is the total variation distance on the space of probability distributions and $C > 0$ is a constant, cf. [25, Theorem 4.9].

Remark 6 (Donsker–Varadhan variational principle)**.** *The transition matrix $\widetilde{P}$ is known to be the unique maximizer of the so called Donsker–Varadhan free-energy functional*

$$\mathcal{F}(Q) \;:=\; \sum_{i,j=1}^N \pi_i^Q\, q_{ij} \log \frac{\Sigma_{ij}}{q_{ij}}, \tag{10}$$

where $Q = (q_{ij})_{i,j=1}^N$ is any irreducible row-stochastic matrix with stationary distribution $\pi^Q = (\pi_i^Q)$, see [13, Section 2], [12, Theorem 3.1.6]. To be more precise, it holds $\mathcal{F}(Q) \leq \mathcal{F}(\widetilde{P}) = \log \lambda_N$. This means that the Markov chain transition matrix $\widetilde{P}$ is the closest (in the above variational sense) to the non–stochastic scale similarity matrix Σ_N. The functional $-\mathcal{F}$ in (10) *can be also interpreted as a stationary relative entropy or Kullback–Leibler divergence.*

Remark 7. *The raw functional values $f(K_j)$ may exhibit multiple local maxima of comparable height, making it ambiguous which scale to select as critical on the basis of $f(K_j)$ alone. The dominant eigenvector v_N of the scale similarity matrix incorporates the* global *filtration structure — weighting each scale j by its coupling to every other scale through the off-diagonal entries Σ_{ij} in*(9) *— and therefore typically refines the partial order carried by $f(K_j)$.*

Remark 8.

Vs.: graph on K_j with a MC on it

3.3 Scale Similarity Matrix as Min or Meet Matrix

Matrices $A = \big(a_{\min(i,j)}\big)_{i,j=1}^{N}$, where $a_1, \dots, a_N$ is a non–negative sequence (in our case: $a_i = f(K_{\varepsilon_i}(S))$ appearing in (8)), is a well-studied class known as *min matrices* or, more generally, *meet matrices on posets* ($A = \big(f(x_i \wedge x_j)\big)_{i,j=1}^{N}$ on a meet-semilattice) [2, 19]. This general theory encompasses e.g. *GCD matrices* $\big(f(\gcd(i,j))\big)$ since the greatest common divisor gcd is the meet on the divisibility poset [34].

Min matrices have a beautiful staircase structure. In this section, we summarize some known facts about them and prove a new one on comonotonicity of their Perron eigenvector (cf. Proposition 6).

3.3.1 Factorisation and Determinant

Define the lower-triangular matrix

$$L_{ij} = \begin{cases} 1, & \text{if } j \le i, \\ 0, & \text{otherwise,} \end{cases}$$

and let $D = \operatorname{diag}(d_1, \dots, d_N)$ with $d_1 = a_1, \quad d_k = a_k - a_{k-1}, \quad k \ge 2$. Then

$$(LDL^{\top})_{ij} = \sum_{k=1}^{i} \sum_{l=1}^{j} d_k \, \delta_{kl} = \sum_{k=1}^{\min(i,j)} d_k = a_{\min(i,j)} = A_{ij},$$

giving the core identity [27, Theorem 5.1]

$$A = L \, D \, L^{\top}. \tag{11}$$

This immediately yields the determinant [19, Corollary 3]:

$$\det A = a_1 \cdot (a_2 - a_1) \cdot (a_3 - a_2) \cdots (a_N - a_{N-1}).$$

From the diagonal D, the matrix A [26], [27, Theorem 8.1] is

- *positive semidefinite* if and only if the sequence $a_1, a_2, \dots, a_N$ is non-negative and non-decreasing;
- *positive definite* if and only if $0 < a_1 < a_2 < \cdots < a_N$.

3.3.2 Local Maxima of the Sequence $\{a_j\}$

Proposition 6 (Comonotonicity of the Perron eigenvector and the sequence $\{a_j\}$)**.** *Let $0 \le a_1 < a_2 < \cdots < a_N$ be a sequence of real numbers and let $A = (A_{ij}) \in \mathbb{R}^{N \times N}$ be the symmetric irreducible matrix with entries*

$$A_{ij} = a_{\min(i,j)}, \qquad 1 \le i, j \le N.$$

Let $v_N = (v_{N1}, v_{N2}, \dots, v_{NN})^{\top}$ denote the positive eigenvector of A associated with the eigenvalue λ_N of largest modulus. Then the vector v_N and the sequence $\{a_j\}_{j=1}^{N}$ are comonotone*: they are increasing, decreasing, and constant at exactly the same indices.*

Proof. By the definition of A, for every index $i \in \{1, \dots, N\}$,

$$(Av_N)_i = \sum_{j=1}^{N} A_{ij} \, v_{Nj} = \sum_{j=1}^{i} a_j \, v_{Nj} + a_i \sum_{j=i+1}^{N} v_{Nj}.$$

The eigenvalue equation $(Av_N)_i = \lambda_N\, v_{Ni}$ then reads

$$\sum_{j=1}^{i} a_j\, v_{Nj} + a_i \sum_{j=i+1}^{N} v_{Nj} = \lambda_N\, v_{Ni}, \qquad 1 \le i \le N. \tag{12}$$

Subtracting equation (12) at index i from the same equation at index $i+1$ yields

$$\begin{aligned}(Av_N)_{i+1} - (Av_N)_i &= \sum_{j=1}^{i+1} a_j\, v_{Nj} + a_{i+1} \sum_{j=i+2}^{N} v_{Nj} - \sum_{j=1}^{i} a_j\, v_{Nj} - a_i \sum_{j=i+1}^{N} v_{Nj} \\ &= a_{i+1}\, v_{N,i+1} + a_{i+1} \sum_{j=i+2}^{N} v_{Nj} - a_i \sum_{j=i+1}^{N} v_{Nj} \\ &= a_{i+1} \sum_{j=i+1}^{N} v_{Nj} - a_i \sum_{j=i+1}^{N} v_{Nj} = (a_{i+1} - a_i) \sum_{j=i+1}^{N} v_{Nj}.\end{aligned}$$

Combined with the right-hand side $(Av_N)_{i+1} - (Av_N)_i = \lambda_N(v_{N,i+1} - v_{Ni})$, this yields the *fundamental recurrence*

$$\lambda_N\,(v_{N,i+1} - v_{Ni}) = d_{i+1}\, S_{i+1}, \qquad S_{i+1} := \sum_{j=i+1}^{N} v_{Nj}, \tag{13}$$

valid for $1 \le i \le N-1$.

By Perron–Frobenius theorem, $v_{Nj} > 0$ for all j, and $\lambda_N > 0$. Consequently, the tail sum S_{i+1} is strictly positive for every $i \in \{1, \dots, N-1\}$. Dividing (13) by $\lambda_N S_{i+1} > 0$ gives

$$\operatorname{sign}(v_{N,i+1} - v_{Ni}) = \operatorname{sign}(d_{i+1}) = \operatorname{sign}(a_{i+1} - a_i), \qquad 1 \le i \le N-1. \tag{14}$$

The sequences $\{v_{Ni}\}_{i=1}^{N}$ and $\{a_i\}_{i=1}^{N}$ therefore are *comonotone*.

□

In particular, v_N and $\{a_j\}$ have identical positions of local maxima, local minima, monotone segments, and boundary extrema. However, the positions of their global maxima may differ.

Remark 9. *The global maxima of coordinates of v_N and the sequence $\{a_j\}$ do not necessarily coincide. The mechanism is clear from the identity*

$$v_{Nm} - v_{Nj} = \frac{1}{\lambda_N} \sum_{k=j+1}^{m} d_k\, S_k, \tag{15}$$

which obviously generalizes relation (13)*. While $a_m - a_j = \sum_{k=j+1}^{m} d_k$, the eigenvector version weights each d_k by S_k. Since S_k is decreasing in k (a tail sum of positive terms), earlier differences are amplified more, creating a systematic bias.*

Proposition 7 (Robustness to perturbations)**.** *It holds*

$$\delta\lambda_j \;=\; \mathbf{v}_j^{\top}\, L\,(\delta D)\, L^{\top}\, \mathbf{v}_j \;=\; \sum_k \delta d_k\, (L^{\top}\mathbf{v}_j)_k^2,$$

where $\mathbf{v}_j$ is the j-th eigenvector of A. Each d_k influences λ_j proportionally to the squared weight $(L^{\top}\mathbf{u}_j)_k^2$ — a cumulative sum of the eigenvector entries.

Proof. Since $A = LDL^{\top}$ and D depends linearly on the a_k, perturbations $a_k \to a_k + \varepsilon_k$ produce

$$\delta A \;=\; L\,(\delta D)\, L^{\top}, \qquad \delta D \;=\; \operatorname{diag}(\delta d_1, \dots, \delta d_N), \quad \delta d_k \;=\; \delta a_k - \delta a_{k-1}.$$

□

formulate correctly and prove

3.3.3 Tridiagonal Inverse

When A is invertible (i.e. all differences $d_k \neq 0$), the inverse A^{-1} is *tridiagonal*. Its entries can be written explicitly in terms d_k [27, Theorem 7.1]:

Diagonal entries are

$$\left(A^{-1}\right)_{ii} = \begin{cases} \dfrac{1}{d_i} + \dfrac{1}{d_{i+1}} & \text{for } 1 \le i < N, \\ \dfrac{1}{d_N} & \text{for } i = N. \end{cases}$$

Off-diagonal entries (superdiagonal and subdiagonal) are

$$\left(A^{-1}\right)_{i,\,i+1} = \left(A^{-1}\right)_{i+1,\,i} = -\frac{1}{d_{i+1}}, \qquad 1 \le i < N.$$

All other entries are zero. Indeed, relation (11) yields $A^{-1} = (L^{-1})^\top D^{-1} L^{-1}$. The key observation is that L^{-1} is *bidiagonal*: it has 1s on the diagonal and -1s on the subdiagonal. Multiplying $(L^{-1})^\top D^{-1} L^{-1}$ then produces the tridiagonal form above.

In terms of the original sequence $\{a_k\}$, A^{-1} has the following explicit form:

$$\begin{pmatrix} \dfrac{1}{a_1} + \dfrac{1}{a_2 - a_1} & -\dfrac{1}{a_2 - a_1} & 0 & \cdots & 0 \\ -\dfrac{1}{a_2 - a_1} & \dfrac{1}{a_2 - a_1} + \dfrac{1}{a_3 - a_2} & -\dfrac{1}{a_3 - a_2} & \cdots & 0 \\ 0 & -\dfrac{1}{a_3 - a_2} & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \dfrac{1}{a_{N-1} - a_{N-2}} + \dfrac{1}{a_N - a_{N-1}} & -\dfrac{1}{a_N - a_{N-1}} \\ 0 & \cdots & \cdots & -\dfrac{1}{a_N - a_{N-1}} & \dfrac{1}{a_N - a_{N-1}} \end{pmatrix}.$$

3.4 Scale Selection for Monotone Functionals f via Embedded Birth–Death Markov Chains

Let $0 < a_1 < a_2 < \cdots < a_N$ be a monotonically increasing sequence of real numbers, for instance, $a_i = f(K_i) f$, $i = 1, \ldots, N$ are values of a functional f on filtration of simplicial complexes K_i. Recall that $d_1 = a_1 > 0$, $d_i = a_i - a_{i-1} > 0$, $i = 2, \ldots, N$ are the consecutive gaps. Then the scale similarity matrix $A = (a_{\min\{i,j\}})_{i,j=1}^N$ built from these scales admits a tridiagonal inverse given in Section 3.3.3. Define the discrete Laplacian

$$\mathcal{L} = \frac{1}{d_1} diag(1, 0, 0, \ldots, 0) - A^{-1},$$

which has negative diagonal entries, positive off-diagonal entries $1/d_{i+1}$ connecting neighbours $i \leftrightarrow i+1$, and row sums that vanish for interior states. The reason to get rid of the part d_1^{-1} in the left topmost entry of $\mathcal{L}$ is that it tells nothing about differences of scales and thus can be omitted. Consequently, $\mathcal{L}$ acts as the generator of a continuous-time birth–death chain on the state space $\{1, \ldots, N\}$ with transition rates

$$q_{i \to i+1} = \frac{1}{d_{i+1}} = \frac{1}{a_{i+1} - a_i}, \qquad q_{i \to i-1} = \frac{1}{d_i} = \frac{1}{a_i - a_{i-1}}, \quad i = 2, \ldots, N-1,$$

so that the birth and death rates are inversely proportional to the gaps between consecutive scales: transitions are fast where the a_k are closely spaced and slow where they are widely spaced. At the boundaries $i = 1$ and $i = N$ the chain is subject to *reflection conditions*: state 1 only admits a birth rate $q_{1 \to 2} = 1/d_2$, and state N only admits a death rate $q_{N \to N-1} = 1/d_N$.

3.4.1 Embedded discrete-time chain and its stationary distribution

The embedded (jump) chain of the above continuous-time process is the discrete-time Markov chain on $\{1, \ldots, N\}$ with transition probabilities

$$P^{\mathrm{BD}}_{i,i+1} = \frac{d_i}{d_i + d_{i+1}}, \qquad P^{\mathrm{BD}}_{i,i-1} = \frac{d_{i+1}}{d_i + d_{i+1}}, \qquad i = 2, \ldots, N-1$$

and boundary modifications $P^{\mathrm{BD}}_{1,2} = P^{\mathrm{BD}}_{N,N-1} = 1$ enforcing reflection at the extremes. It is irreducible and periodic with period 2, due to its birth-and-death nature as well as reflections at the boundary. Since the chain is finite and irreducible, it is positive recurrent and has a unique stationary distribution.

Let r_i denote the total outgoing rate from state i in the continuous-time chain, i.e.,

$$r_i = \begin{cases} \dfrac{1}{d_2}, & i = 1, \\ \dfrac{1}{d_i} + \dfrac{1}{d_{i+1}}, & i = 2, \ldots, N-1, \\ \dfrac{1}{d_N}, & i = N. \end{cases}$$

The detailed-balance equation for the embedded chain reads

$$\pi^{\mathrm{BD}}_i \, P^{\mathrm{BD}}_{i,i+1} = \pi^{\mathrm{BD}}_{i+1} \, P^{\mathrm{BD}}_{i+1,i}$$

for all i and, after substitution and telescoping, yields

$$\frac{\pi^{\mathrm{BD}}_{i+1}}{\pi^{\mathrm{BD}}_i} = \frac{r_{i+1}}{r_i},$$

so that π^{BD}_i is proportional to r_i: $\pi^{\mathrm{BD}}_i \propto r_i$. The normalised stationary distribution of the embedded birth–death chain is therefore

$$\pi^{\mathrm{BD}}_i = r_i / \bar{r}, \qquad i = 2, \ldots, N-1 \tag{16}$$

with the boundary modifications $\pi^{\mathrm{BD}}_1 = (\bar{r} d_2)^{-1}$ and $\pi^{\mathrm{BD}}_N = (\bar{r} d_N)^{-1}$, where $\bar{r} = \sum_{j=1}^{N} r_j$.

Although ordinary n-step convergence fails in the periodic case, the Cesàro averages converge:

$$\frac{1}{n} \sum_{k=0}^{n-1} P^{BD,k}(i,j) \longrightarrow \pi^{\mathrm{BD}}_j, \qquad n \to \infty$$

for all $i, j = 1, \ldots, N$, where $P^{BD,k}(i,j)$ is the probability for the embedded Markov chain to reach state j from state i in k steps. This follows from the ergodic theorem for Markov chains [30, Theorem 1.10.2] appying it to the indicator function that the chain is at state j and then using the L^1–convergence by Lebesgue theorem.

3.4.2 Identification of the critical scales

Since $\pi^{\mathrm{BD}}_i \propto r_i$ for all i, the stationary distribution is determined entirely by the local increments of the complexity functional f around state i. It is *small* exactly at those indices i where both d_i and d_{i+1} are *large*, i.e., where the functional f undergoes sharp jumps on *both* sides of the scale K_i; it is *large* at indices where f varies slowly, i.e., where the filtration crosses a plateau of nearly constant complexity.

Locally large jumps of f signal a *topological transition* in the filtration. The critical scales are therefore naturally identified as

$$I_{crit} = \operatorname*{arg\,min}_{i=1,\ldots,N} \pi^{\mathrm{BD}}_i, \tag{17}$$

i.e. as a set of states in which the continuous–time birth-and-death Markov chain spends most of its time. The apparent paradox — argmin of a stationary distribution identifying a critical scale — is resolved once one observes that the discrete chain is constructed from the inverse A^{-1} of the scale similarity matrix, so that states with low π_i^{BD} are precisely those connected to their neighbours by slow transition rates.

This criterion has a direct interpretation at the level of the complexity functional f, independently of any spectral analogy: for $i^\star \in I_{crit}$, ansatz (17) selects the filtration step at which the pair of increments $d_{i^\star}, d_{i^\star+1}$ is simultaneously largest, i.e., the step across which the functional f exhibits its most pronounced local variation on both sides. Such a step separates two regions of the filtration on which f varies slowly — plateaux of approximately constant topological complexity — and therefore corresponds to the most informative scale at which to analyse the complex. The recipe is a topological analogue of the classical *elbow rule* for scree plots in PCA, with the role of eigenvalue gaps played by jumps of the complexity functional f along the filtration.

Remark 10 (Decreasing sequences)**.** *If the input sequence $\{a_j\}_{j=1}^N$ is positive and monotonically* decreasing *rather than increasing, all of the above applies verbatim after replacing the differences d_j by their absolute values; equivalently, one multiplies each d_j by -1, which leaves the rates $|d_j|^{-1}$, the embedded transition probabilities, the stationary distribution* (16)*, and the critical-scale identification* (17) *unchanged. Hence the methodology is invariant under reversal of the enumeration of scales.*

4 Examples

In this section, we consider the functionals of a simplical complex K of dimension d of the generic form $f(K) := f_{Top}(K) \cdot f_{Comb}(K)$, where

$$f_{Top}(K) := \delta_{-1} + \sum_{j=0}^{d-1} \delta_j \beta_j^{\alpha_j}(K) + \delta_d \mathrm{cup}_R^{\alpha_d}(K) + \delta_{d+1} \mathrm{PC}^{\alpha_{d+1}}(K), \tag{18}$$

$$\begin{aligned} f_{Comb}(K) := \gamma_{-1} + \sum_{j=0}^{d} \gamma_j f_j^{\xi_j}(K) + \gamma_{d+1} CR_{k,m}^{\xi_{d+1}}(K) + \gamma_{d+2} c_r^{\xi_{d+2}}(K) \\ + \gamma_{d+3} \Delta_t^{\xi_{d+3}}(K) + \gamma_{d+4} \tau^{\xi_{d+4}}(K) + \gamma_{d+5} \nu^{\xi_{d+5}}(K), \end{aligned} \tag{19}$$

and coefficients δ_j, $j = -1, 0, \ldots, d+1$ as well as γ_j, $j = -1, 0, \ldots, d+5$ are some real numbers. Here, the combinatorial and topological features in the linear combinations in (18)-(19) are taken from Sections 2.2 and 2.3. Powers α_j, $j = 0, \ldots, d+1$ and ξ_j, $j = 0, \ldots, d+5$ are real numbers that may also be negative. The product form of f provides flexibility in the choice of their combinations as well as allows for (non)monotone functionals f constructed from monotone f_{Top} and f_{Comb}. The issues of subadditivity and nonnegativity of f are discussed in concrete examples below for a specific choice of simplicial operation $\oplus$ and numerical values of coefficients δ_j and γ_j.

Depending on the context, the operation $\oplus$ on simplicial complexes will be the union, intersection or flagification of their union.

Let us start with the first toy example. It states that the spectral scale methodology leads to the intuitive choice of filtration parameter which one would make also without any serious mathematics.

Example 5 (Monotone functional $f \geq 0$ w.r.t. union operation)**.** *Assume we can observe the positions of a celestial body moving around a star along an elliptical orbit, at equidistant time instants. Can we reconstruct the topology of this orbit using the spectral scale selection?*

Consider the orbit as an ellipse $\mathcal{E}$ *described by the equation* $x^2/a^2 + y^2/b^2 = 1$ *in its ecliptic plane with semi-axes* $a \geq b > 0$. *Take the observation points* $S = \{s_0, \ldots, s_{n-1}\} \subset \mathcal{E}$ *with Euclidean coordinates* $s_j = (x_j, y_j)$ *on this plane which are recorded over the whole orbital period equidistantly in time, compare Figure 1. The star is put to the leftmost of the foci of* $\mathcal{E}$. *The orbital eccentricity is* $e = \sqrt{1 - b^2/a^2}$. *According to Kepler's second law, the line joining the planet to the attracting focus sweeps out equal areas in equal times. Let the full orbital period be* T. *We divide the period into* n *equal time intervals:* $t_j = \frac{jT}{n}$, $j = 0, 1, \ldots, n-1$. *The corresponding mean anomaly is* $M_j = \frac{2\pi}{T} t_j = \frac{2\pi j}{n}$, $j = 0, 1, \ldots, n-1$. *To compute the actual position of the planet, one introduces the* eccentric anomaly E_j *defined implicitly by Kepler's equation*

$$M_j = E_j - e \sin E_j, \qquad j = 0, 1, \ldots, n-1.$$

Put $E_n = E_0 + 2\pi$. *The eccentric anomaly* E_j *is the parameter of the standard ellipse parametrization which yields* $s_j = (x_j, y_j)$ *with*

$$x_j = a \cos E_j, \qquad y_j = b \sin E_j, \qquad j = 0, \ldots, n-1.$$

The geodesic distance from point s_j *to point* s_{j+1} *along the direction of motion is*

$$\rho_j := a \int_{E_{j-1}}^{E_j} \sqrt{1 - e^2 \cos^2 u}\, du, \qquad j = 1, \ldots, n.$$

Consider an even $n = 2N$ *and reorder the distances* ρ_j, $j = 1, \ldots, N$ *in strictly increasing*

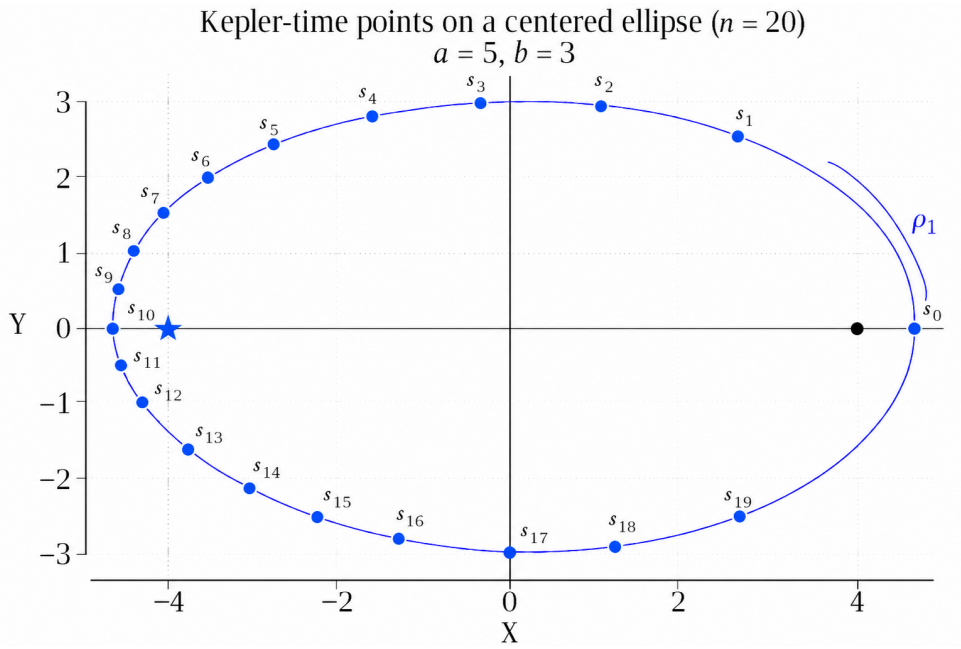


Figure 1: Set S of $n = 20$ orbital points on a centered ellipse with semi-axes $a = 5$ and $b = 3$.

order: $\rho_{(1)} < \ldots < \rho_{(N)}$. *Take a filtration of graphs* K_j, $j = 1, \ldots, N$ *with vertex set* S *where two points* s_k *and* s_l, $|k - l| = 1$ *or* $k = 0$, $l = n - 1$ *from* S *are connected by an edge whenever the geodesic distance between them is smaller than or equal to* $\varepsilon_j = \rho_{(j)}$, $j = 1, \ldots, N$. *Equip* $\{K_j\}$, $j = 1, \ldots, N$ *with the union operation. Consider the functional* $f(K_j) = \beta_0^{-1}(K_j) \geq 0$. *It is of the form* (18) *with* $\delta_0 = 1$ *and* $\alpha_0 = -1$. *All other parameters* δ_j *and* α_j *are set to be zero. It is not difficult to see that* f *is strictly increasing on the chosen filtration with*

$$f(K_j) = (n - 2j)^{-1}, \quad j = 1, \ldots, N-1, \quad f(K_N) = 1.$$

Using the rule (17) *for the critical scale selection, we have to compute differences* $d_j = f(K_j) - f(K_{j-1}) > 0$ *for any* $j = 2, \ldots, N$. *Then, compute* r_j *from* (16) *to get*

$$r_1 = \frac{1}{d_2} = \frac{(n-4)(n-2)}{2}, \quad r_j = \frac{1}{d_j} + \frac{1}{d_{j+1}} = (n-2j)^2, \quad j = 2, \ldots, N-1, \quad r_N = \frac{1}{d_N} = 2.$$

The critical scale number is

$$N = argmin_{j=1,\ldots,N}\, r_j.$$

Thus, $\varepsilon_N = \rho_{(N)} = \rho_1 = \rho_n$ *is the minimal possible scale to reveal the unique connected component in* K_N *being a linear approximation of original elliptical orbit* $\mathcal{E}$.

Example 6 (Non-monotone functional $f \geq 0$ w.r.t. union operation)**.** *functional* $f = f_{Top}(K) \cdot f_{Comb}(K)$ *as a product of two evtl. monotone functionals ?*

Example 7 (Subadditive functional f w.r.t. ? operation)**.** *Subadditive functional* $f =?$ *that can also be negative...*

Example 8 (Monotone functional $f \geq 0$ w.r.t. intersection operation)**.** *functional* $f =?$

Example 9 (General functional f)**.** *"no-properties" functional* $f =?$*, use of matrix* Σ_N^2.

5 Numerical Results

For testing our PCA algorithm on simplicial complexes, consider:

1. **Primary**: Vietoris-Rips (computational efficiency)
2. **Validation**: Čech complex (geometric accuracy)
3. **High Dimensions**: Alpha complex (avoids curse of dimensionality)

Run Algorithm for $Cech_\varepsilon$ vs VR_ε vs $Cech_{2\varepsilon}$ and try 2ε accuracy for VR_ε.

5.1 Data Generation

Reproducibility and Code

For VR and persistence computation we used `GUDHI` [17].

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